

Foundation

Foundation is the lowest part of the building or the civil structure that is in direct contact with the soil which transfers loads from the structure to the soil safely. Generally, the foundation can be classified into two, namely shallow foundation and deep foundation. A shallow foundation transfers the load to a stratum present in a shallow depth. The deep foundation transfers the load to a deeper depth below the ground surface.

Foundations are provided for all load carrying structure for following purposes:

- Foundations are the main reason behind the stability of any structure. The stronger is the foundation, more stable is the structure.
- The proper design and construction of foundations provide a proper surface for the development of the substructure in a proper level and over a firm bed.
- Specially designed foundation helps in avoiding the lateral movements of the supporting material.
- A proper foundation distributes load on to the surface of the bed uniformly. This uniform transfer helps in avoiding unequal settlement of the building. Differential settlement is an undesirable building effect.
- The foundation serves the purpose of completely distributing the load from the structure over a large base area and then to the soil underneath. This load transferred to the soil should be within the allowable bearing capacity of the soil.

Functions of Foundation

Based on the purposes of foundation in construction, the main functions of the foundation can be enlisted as below:

1. Provide overall lateral stability for the structure
2. Foundation serve the function of providing a level surface for the construction of substructure
3. Load Distribution is carried out evenly
4. The load intensity is reduced to be within the safe bearing capacity of the soil
5. The soil movement effect is resisted and prevented
6. Scouring and the undermining issues are solved by the construction of foundation

Requirements of a Good Foundation

The design and the construction of a well-performing foundation must possess some basic requirements that must not be ignored. They are:

1. The design and the construction of the foundation is done such that it can sustain as well as transmit the dead and the imposed loads to the soil. This transfer has to be carried out without resulting in any form of settlement that can result in any form of stability issues for the structure.
2. Differential settlements can be avoided by having a rigid base for the foundation. These issues are more pronounced in areas where the superimposed loads are not uniform in nature.
3. Based on the soil and area it is recommended to have a deeper foundation so that it can guard any form of damage or distress. These are mainly caused due to the problem of shrinkage and swelling because of temperature changes.
4. The location of the foundation chosen must be an area that is not affected or influenced by future works or factors.

What is a Plinth Beam and its Purpose in a Building?

Plinth beam is a reinforced concrete beam constructed between the wall and its foundation. Plinth beam is provided to prevent the extension or propagation of cracks from the foundation into the wall above when the foundation suffers from settlement. Plinth beams distribute the load of the wall over the foundation evenly.

Applications of Plinth Beam

1. It is mandatory to provide plinth beam in areas that are prone to earthquake.
2. Construction of plinth beam above the natural ground is another application of this type of beam.

What is a lintel?

In construction, a lintel is a beam placed across the top of an opening in a wall. Used above empty spaces like entrances and windows in buildings, lintels support the weight of the structures above them. Typically, the beams fit seamlessly into the wall, matching the width of the wall and extending past either end of the opening. Mostly used in brick and masonry

buildings, lintels stem from a variety of materials depending on the structure's support needs, contractor preference and building aesthetic. Similar to an arch but flat and easier to install, lintels can be an important structural element in a building.

Where are lintels used?

Lintels support the weight of structures above wall openings, doorways and windows in primarily brick and masonry buildings. Open spaces in walls often lack structural integrity because there is nothing above the gap to support the load on top of the opening. To combat this, lintel beams extend past either side of the doorway or opening. Builders install them into the wall to disperse the weight of materials directly atop the open space.

Chejja

Many of us may not be familiar with the term Chajja. ‘Chajja’ is the projected element just above the window, which shades the opening, stops rain from entering the room and reduces sky glare while looking out of the room. It is also known as “Sun-Shade” that is a sloping or horizontal structural overhang, usually provided for protection from sun and rain or for architectural considerations at lintel level.

Importance of Chajja

1. Usually protects from external sunlight.
2. Also protects from rainwater entering into the building.
3. It also serves as aesthetic decoration with little design on it from architectural point of view.
4. Windows-Chajja will not let water drip from above and soak the woods of windows. Otherwise the wood will absorb water and expand.
5. Can be used as a support for AC.

What is beam structure?

A beam structure, sometimes simply referred to as a beam, is a type of structure used in construction and engineering to provide a safe and efficient load path that effectively distributes weight throughout the foundation of a building. These beams support the load by resisting being bent under the load's pressure. Beams resist this force in a lateral way as the force is applied to the axis.

The primary purposes of beam structure include:

1. Offsetting shear forces and/or beam momentum
2. Resisting loads
3. Distributing loads in a uniform fashion
4. Uniting the structure together

Columns

Columns in civil engineering can be defined as vertical structural elements that act as supports and primarily support axial compressive loads. They are slender members designed as a support to hold the ceiling and roof, and the weight acting on them. The load acting on the structure is transmitted to the foundation through these supports.

Based on Loading

1. **Axially Loaded Columns:-** Axially loaded columns are used when the compressive vertical load acts along the centroidal axis of the column. These columns do not resist bending moments as such and coinciding vertical compressive loads along the centre of gravity of the column is not practical, so they are not used commonly.
2. **Eccentrically Loaded Uniaxial Column:-** When the load acts at any other point apart from the centroid of the column's cross-section, this type of loading is called eccentric loading. When the loads act eccentrically on any one of either X or Y-axis of the column, then the columns are termed as eccentrically loaded uniaxial columns.

Slabs

Slab is an important structural element which is constructed to create flat and useful surfaces such as floors, roofs, and ceilings. It is a horizontal structural component, with top and bottom surfaces parallel or near so. Commonly, slabs are supported by beams, columns (concrete or steel), walls, or the ground. The depth of a concrete slab floor is very small compared to its span.

Types of Loads on a Slab

Types of loads acting on a slab include:

1. Dead load of the slab
2. Live load
3. Floor finish load
4. Snow load in the case of roof slab
5. Earthquake loads

Staircase

Staircase is an important component of a building providing access to different floors and roof of the building. It consists of a flight of steps and one or more intermediate landing slabs between the floor levels.

Stairs can be defined as series of steps suitably arranged for the purpose of connecting different floors of a building. It may also be defined as an arrangement of treads, risers, stringers, newel post, hand rails, and baluster, so designed and constructed as to provide an easy and quick access to the different floors.

A set of steps formed to make it possible to pass to another level on foot by putting one foot after the other an alternate steps to climb up or down the stair. Stairs can be made of concrete, stone, wood, steel or combination of any of these.

Primary Functions

1. Provide an access from one floor to another.
2. Provide a safe means of travel between floors.
3. Provide a degree of insulation where part of a separating element between compartments in a building.
4. Provide a suitable means of escape in case of fire.
5. Provide a mean of conveying fittings and furniture between floor levels.